1 Diagrams and summary charts for Problems 11–12

The following diagrams summarize the geometric content of the clutching construction, the unique nontrivial bundle over S^1, and the associated sphere bundles. They are intended to support the solutions of Problems 11–12 by showing how the abstract gluing data is represented geometrically.

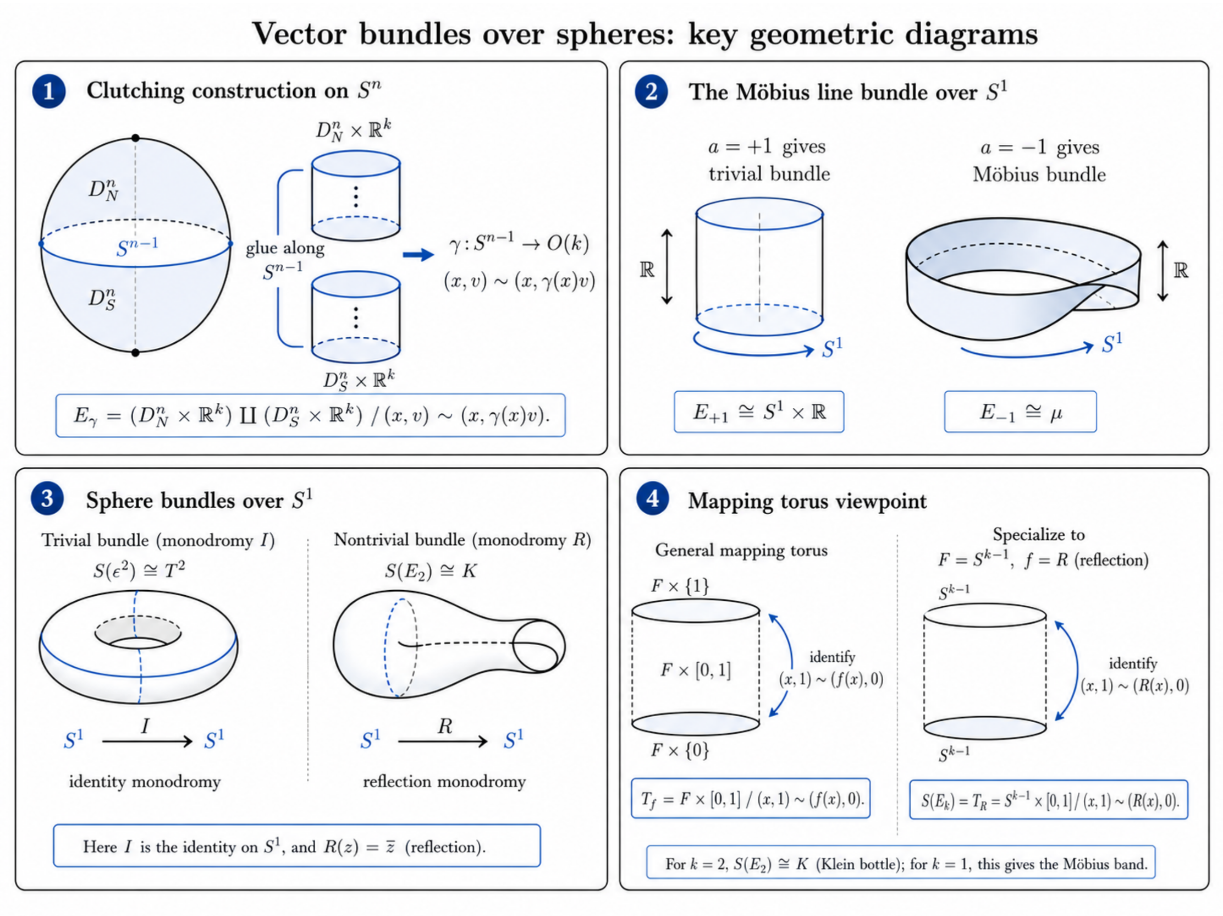

Figure 1: Key geometric diagrams for vector bundles over spheres. The first panel shows the clutching construction over S^n; the second compares the trivial line bundle over S^1 with the Möbius line bundle; the third compares the torus and Klein bottle as sphere bundles over S^1; the fourth interprets these examples as mapping tori.

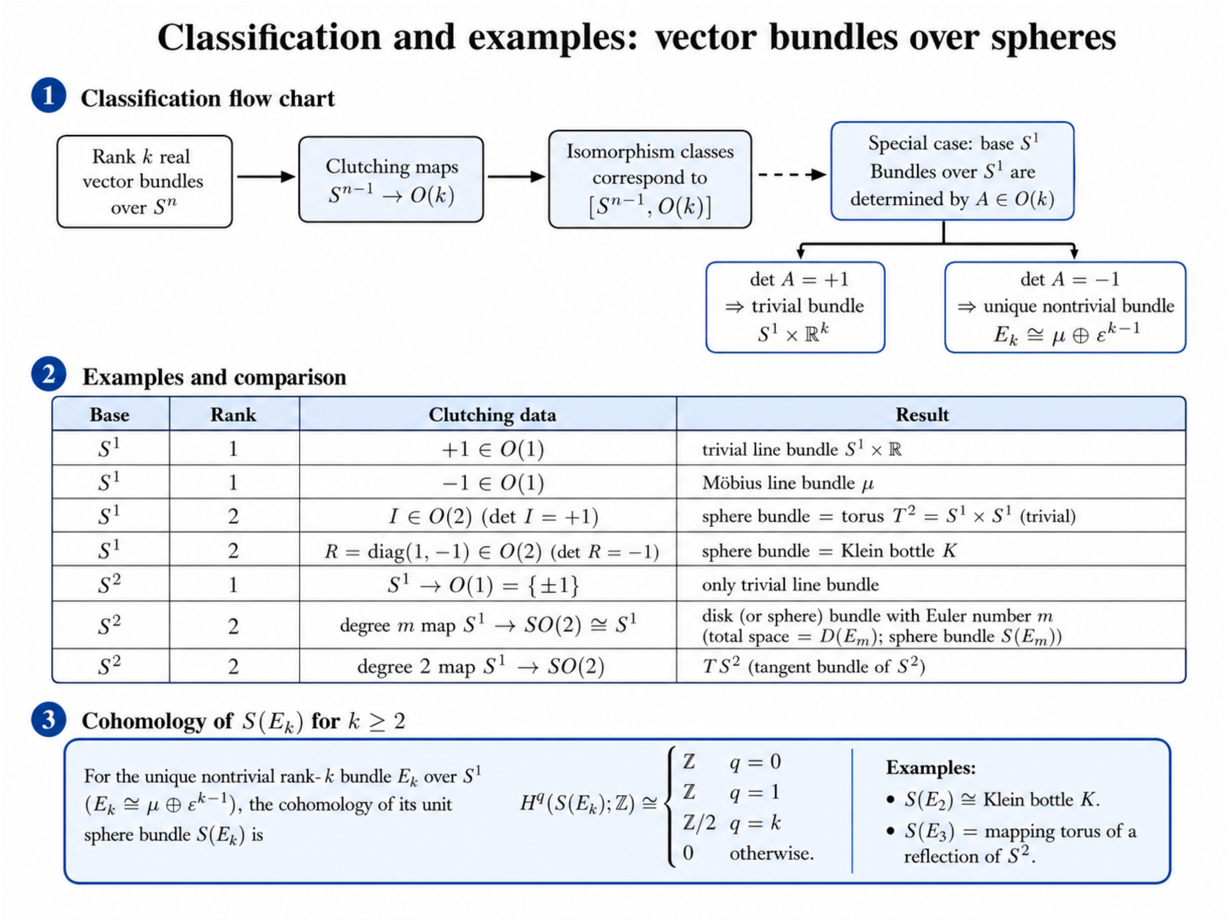

Figure 2: Classification chart and examples for vector bundles over spheres. Rank k real vector bundles over S^n are described by clutching maps $S^{n-1} \to O(k)$. The special case of bundles over S^1 reduces to the determinant of the monodromy matrix $A \in O(k)$. The chart also records the cohomology of the unit sphere bundle $S(E_k)$ for the unique nontrivial bundle $E_k \to S^1$.

How to read the diagrams

In Figure 1, the bundle $E_\gamma \to S^n$ is obtained by gluing two trivial bundles over the northern and southern hemispheres. The gluing is performed along the equator S^{n-1}, and the gluing rule is encoded by the clutching map

$$\gamma : S^{n-1} \longrightarrow O(k).$$

Thus the whole bundle is determined by how the fibers are identified along the equator.

For bundles over S^1, the monodromy matrix $A \in O(k)$ determines the bundle. If $\det A = +1$, the bundle is trivial. If $\det A = -1$, one obtains the unique nontrivial rank k bundle

$$E_k \cong \mu \oplus \varepsilon^{k-1}.$$

Its unit sphere bundle is the mapping torus of a reflection of S^{k-1}:

$$S(E_k) \cong S^{k-1} \times [0,1]/(u,1) \sim (Ru, 0).$$

For $k = 2$, this gives the Klein bottle:

$$S(E_2) \cong K.$$